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Supplementary information

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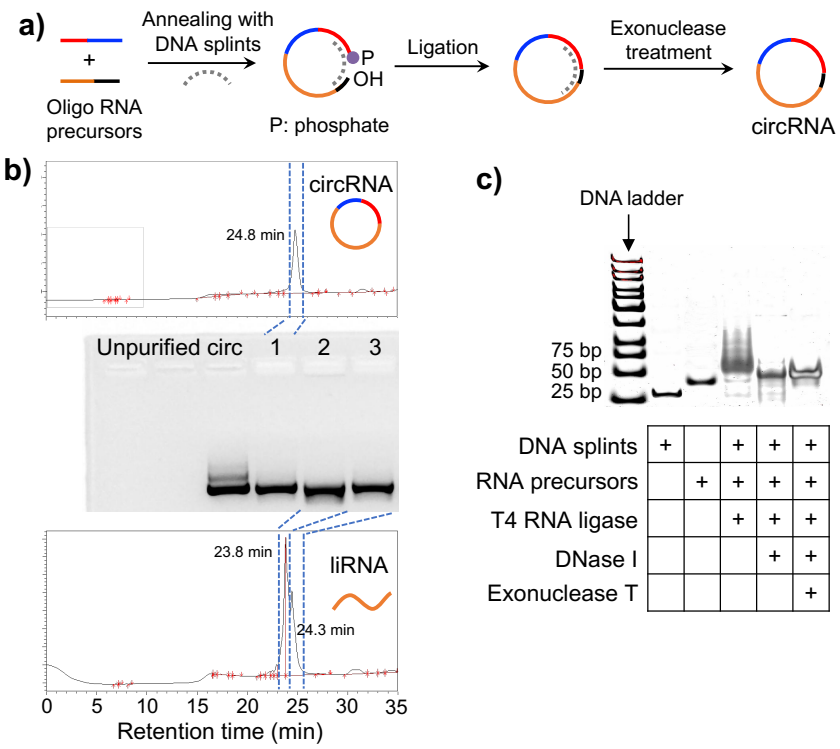
# Small circular RNAs as vaccines for cancer immunotherapy

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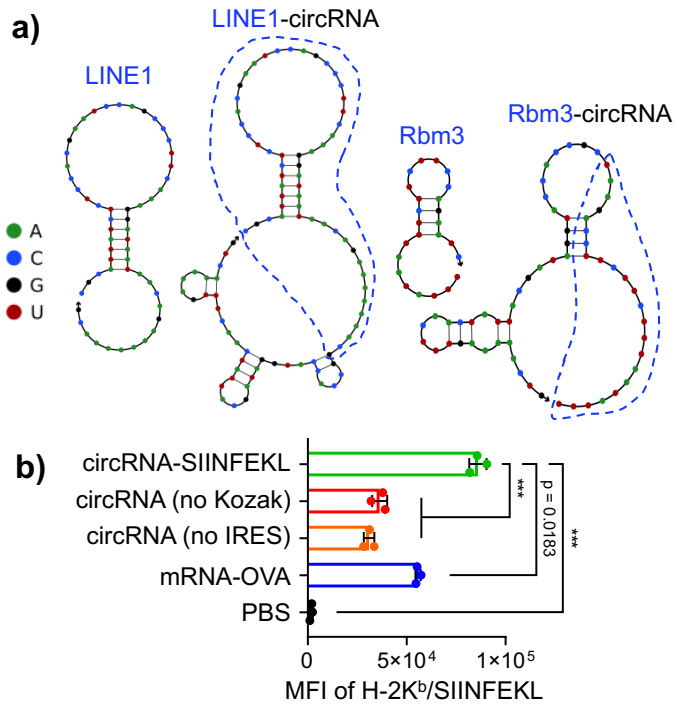
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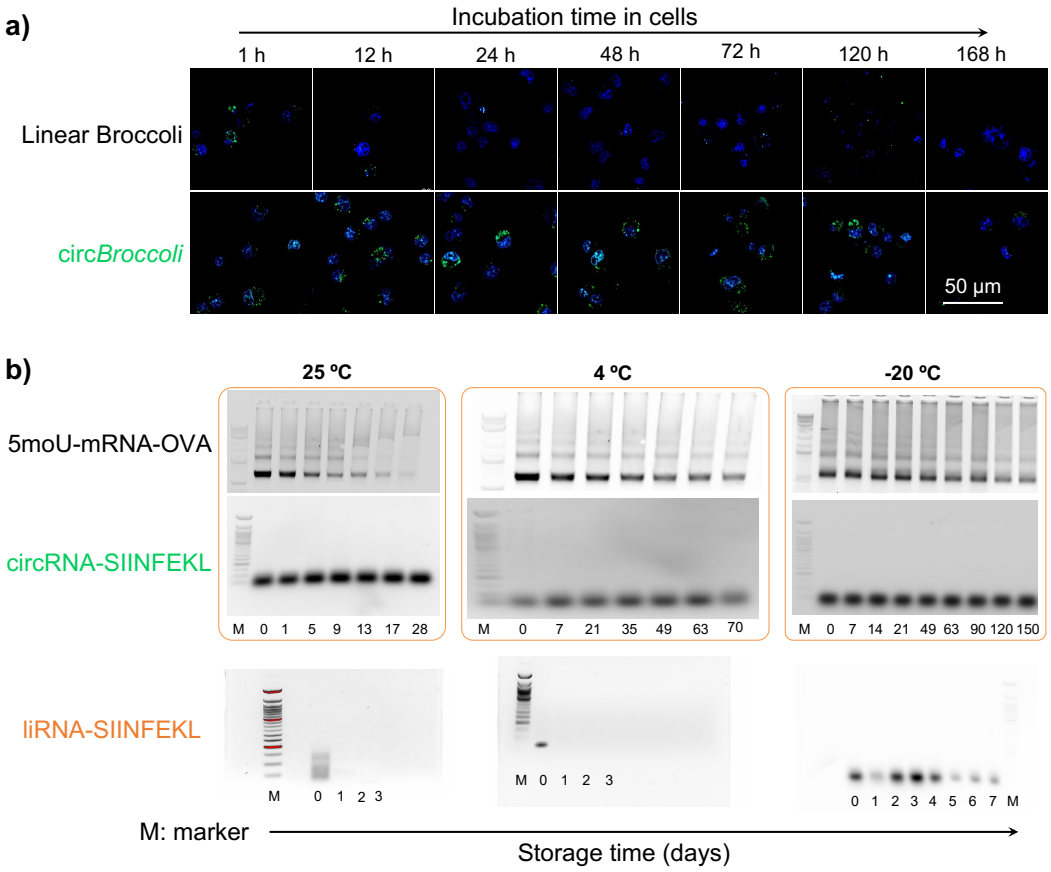
**Supplementary Table 1 l** RNA sequences.
**Supplementary Table 2 l** Structures of circRNA-SIINFEKL, circRNA-OVA, and OVA-/SIINFEKL- encoding mRNA.
**Supplementary Table 3 l** A list of antibodies used in this study.
**Supplementary Table 4 l** Peptide sequences.



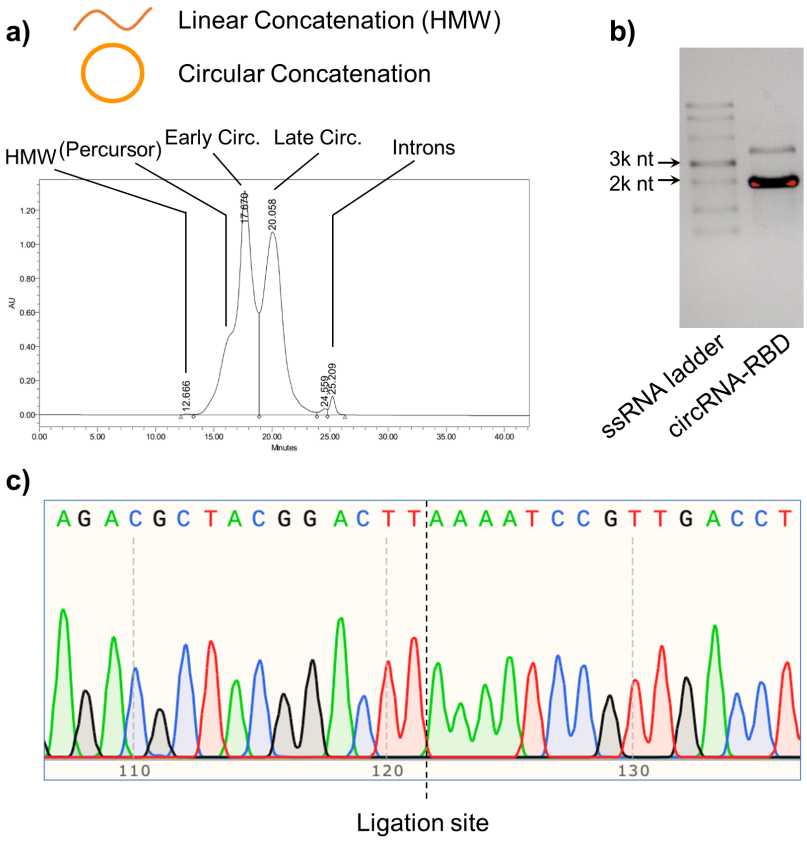

**Supplementary Fig. 1 l Validation of small circRNA synthesis. a**, Schematic illustration of small circRNA synthesis. **b,** HPLC and gel electrophoresis verified the purity of circRNA, which showed a different retention time than that of linear RNA. The corresponding fractions of HPLC samples were collected and subject to agarose gel electrophoresis. **c**, Agarose gel electrophoresis validated the synthesis of small circRNA. Even though ligation-based circRNA synthesis could result in multimeric circRNA, such byproducts were not detectable under this experiment condition. This is likely due to the relatively low concentration and the favored intramolecular ligation relative to intermolecular ligation, the latter forming multimeric circRNA.



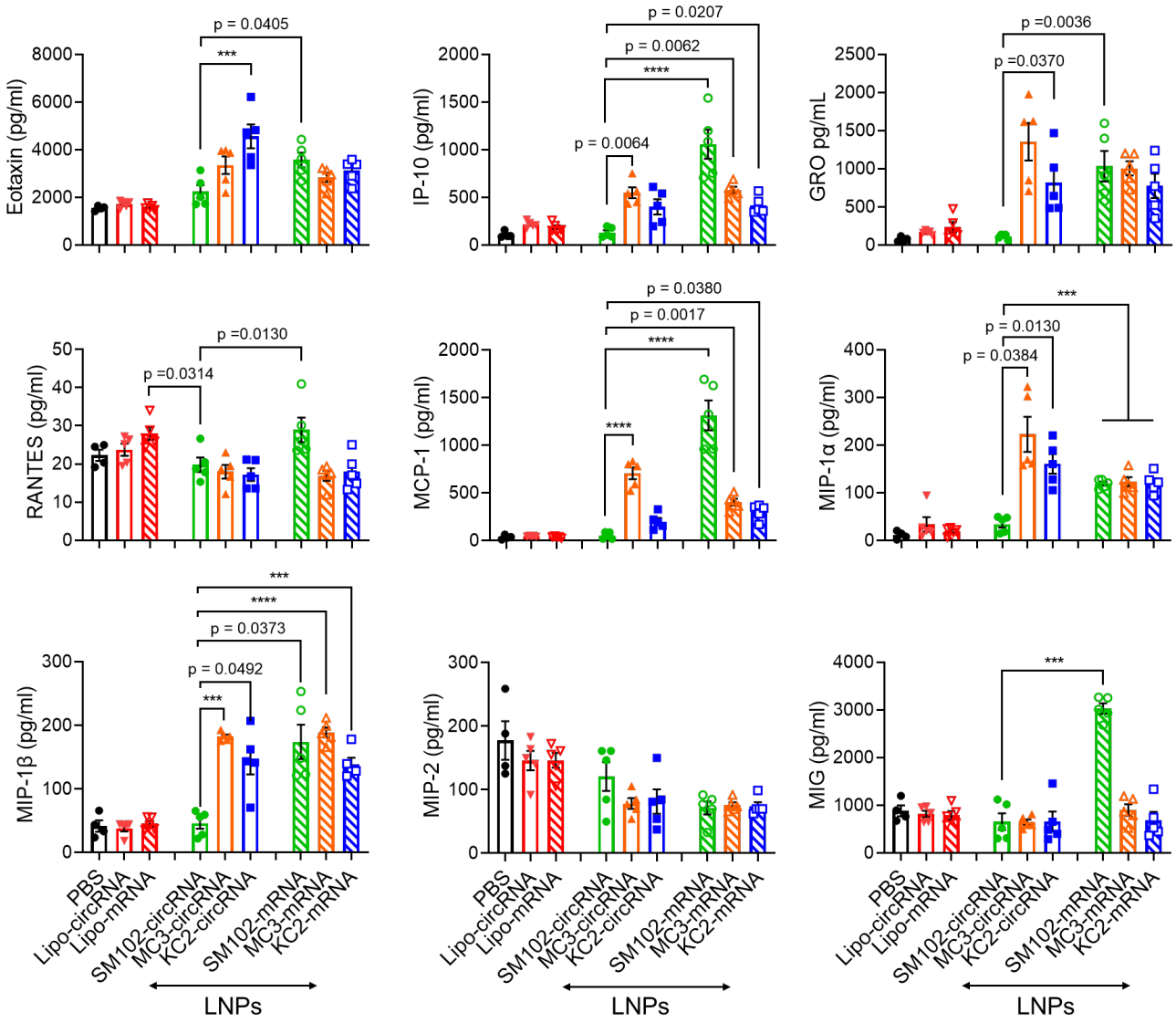

**Supplementary Fig. 2 | *In vitro* antigen presentation by DCs treated with small circRNA vaccine. a**, the secondary structures (predicted by NUPACK) of IRES LINE1 and Rbm3 (left) and the corresponding circRNA-SIINFEKL (right). Blue boxes indicate IRES domains. **b**, Flow cytometric analysis of SIINFEKL/H-2K$^b$ complex level on DC2.4 cells. DC2.4 cells were transfected with various SIINFEKL or OVA vaccines for 24 h, and were then stained with an APC-conjugated H-2K$^b$/SIINFEKL antibody followed by flow cytometric analysis. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, by one-way ANOVA with Bonferroni post-test.



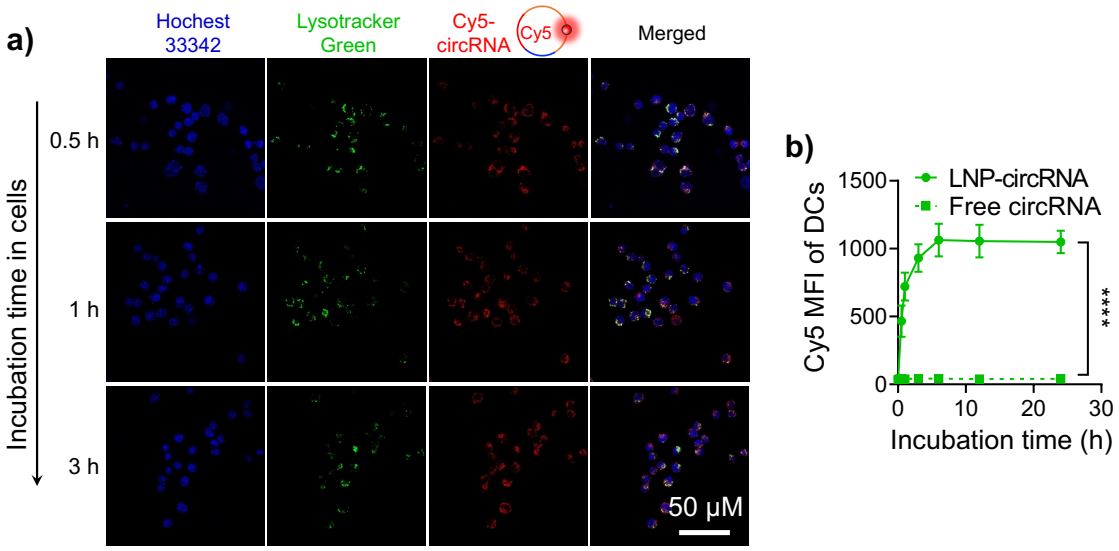

**Supplementary Fig. 3 l High stability of small circRNA relative to long modified mRNA and large circRNA. a**, confocal microscopy images (**green: Broccoli-DFHBI-1T**; blue: nuclei) of DC2.4 cells treated with linear or circular *Broccoli* for 1-168 h, prior to adding DFHBI-1T before analysis. **b**, Representative agarose gel electrophoresis images of small circRNA-SIINFEKL, liRNA-SIINFEKL, and 5moU-modified CleanCap mRNA-OVA after storage in PBS for the indicated number of days.




**Supplementary Fig. 4 l Validation of circRNA-RBD synthesis. a**, HPLC chromatogram of circRNA-RBD product after column purification. **b,** an agarose gel electrophoresis image showing the successful synthesis of circRNA-RBD (marked by arrow). circRNA-RBD product was treated with RNase R and column purified before gel electrophoresis. The >3k-nt band is presumably dimeric circRNA. **c**, Sanger sequencing of circRNA-RBD cDNA indicates the precise circularization circRNA-RBD.




**Supplementary Fig. 5 l** Serum concentrations of chemokines 12 h after administration of 5 $\mu$g circRNA-SIINFEKL and 5moU-mRNA-OVA, each of which was formulated in liposome, SM-102 LNPs, MC3 LNPs, and KC2 LNPs, respectively (n = 5). Ns: non-significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, by one-way ANOVA with Bonferroni post-test.

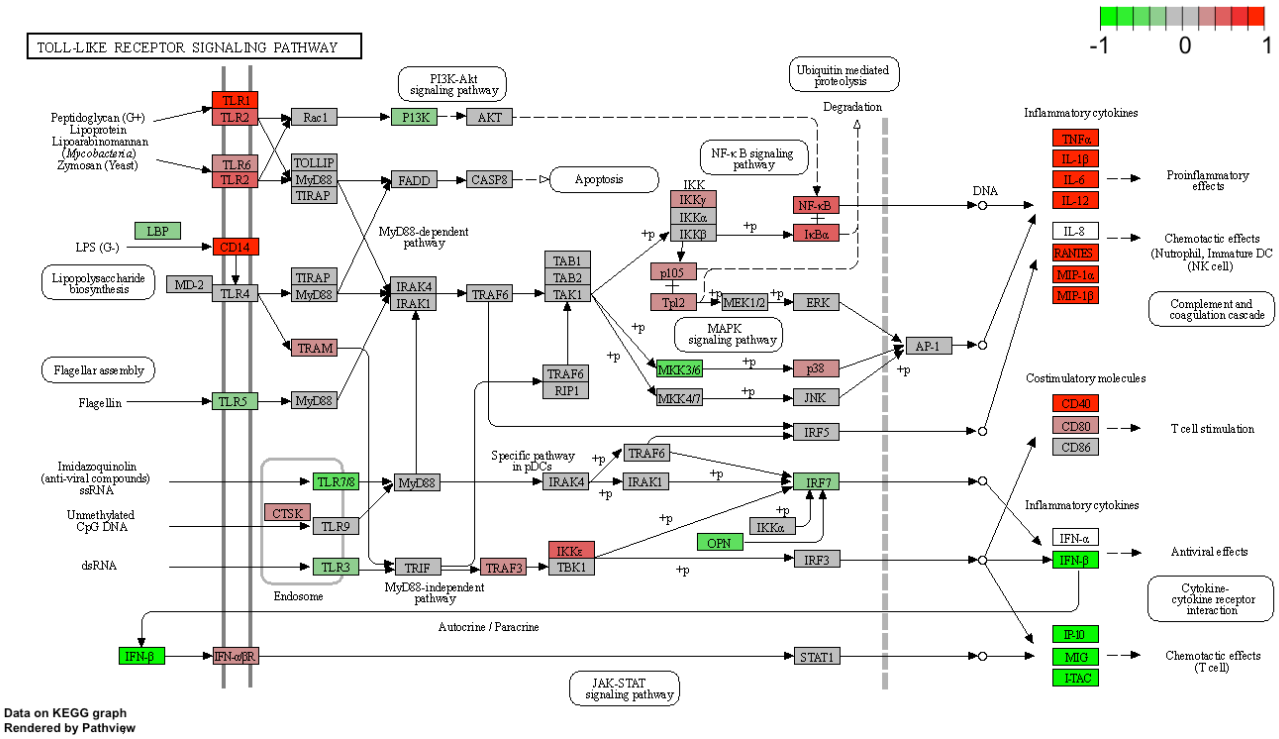



**Supplementary Fig. 6 l LNPs promoted the intracellular delivery of circRNA in DCs. a**, Confocal microscopy images showing efficient LNP-circRNA-Cy5 delivery to DC2.4 cells (100 nM circRNA). Cy5 signal from circRNA not colocalized with Lysotracker indicates circRNA endosome escape. Blue: nuclei stained with Hoechst33342. Green: endolysosome stained with LysoTracker Green DND-26. Red: Cy5-circRNA. **b**, Flow cytometric analysis (MFI) of DC2.4 cells incubated with free Cy5-circRNA or LNP-Cy5-circRNA showed that LNPs enhance circRNA delivery to DCs. LNPs were prepared using SM-102 as the ionizable lipid. ****$p < 0.0001$, by t-test of the AUC.

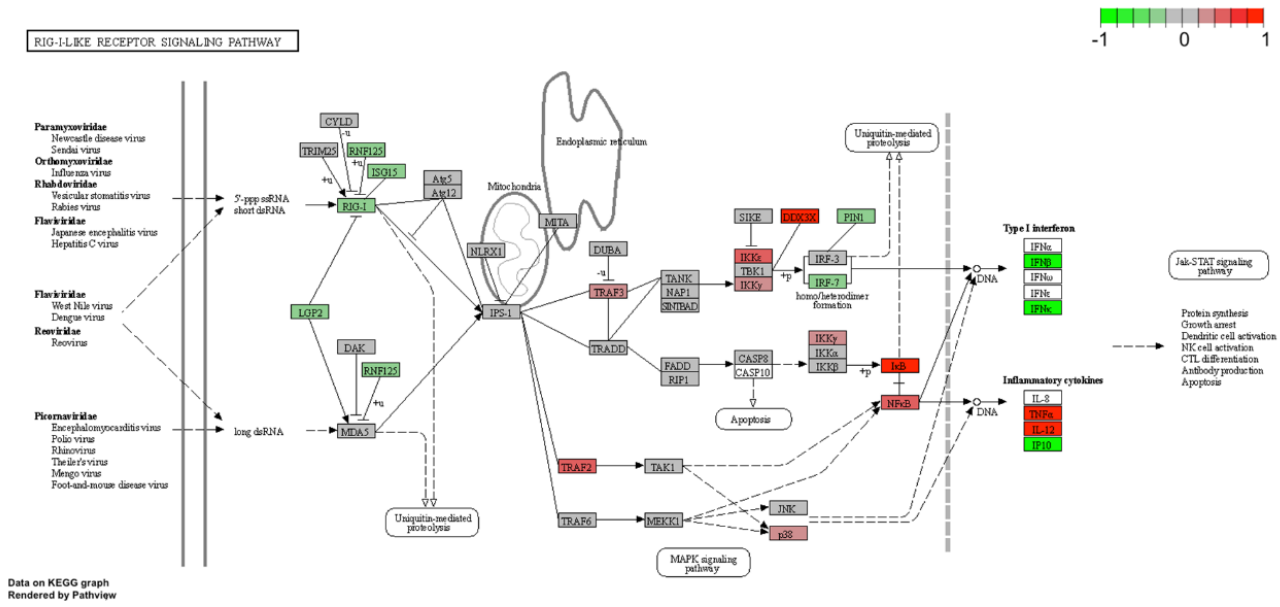
We also studied circRNA delivery by liposome (lipo-circRNA) synthesized from 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP), 1,2-distearoyl-sn-glycero-3-PE (DSPE)-PEG2000, and cholesterol (molar ratio: 10: 1: 10)[S1] that has great safety. Dynamic light scattering (DLS) showed the lipo-circRNA diameter of ~120 nm in PBS, with only slight size increase in 1% FBS (**Extended Data Fig. 1a**), suggesting good stability. We loaded SIINFEKL-encoding circRNA into liposomes at N:P ratios of 1:1, 3:1, and 10:1, respectively, (**Extended Data Fig. 1b**) all of which had ~95% circRNA encapsulation efficiency (amine (N) in liposome, phosphate (P) in RNA). As evaluated by the SIINFEKL presentation on DCs, N:P ratio of 3:1 and 10:1 resulted in most SIINFEKL presentation on DCs (**Extended Data Fig. 1c-d**). We chose 10:1 for further study due to the similarity to Lipofectamine.

DCs are professional APCs that mediate antigen presentation to elicit adaptive immunity. To study intracellular lipo-circRNA delivery to DCs, circRNA was labeled with a Cy5-complementary DNA (cDNA) for fluorescence tracking of intracellular circRNA. After incubation of DCs with liposomal *vs.* free circRNA-Cy5 for a series of time up to 24 h, flow cytometric analysis revealed efficient cell uptake of lipo-circRNA within 1 h, in contrast to undetectable cell uptake of free circRNA (**Extended Data Fig. 1f-g**). To further explore cellular localization of circRNA, the nuclei and endolysosomes of above-treated DCs were also stained with Hoechst33342 and LysoTracker-Green, respectively. Fluorescence confocal laser scanning microscopy (CLSM) verified the efficient cell uptake of lipo-circRNA into endolysosomes, and importantly, liposomes mediated efficient escape of circRNA from the endolysosome to cytosol where peptide antigens are translated (**Extended Data Fig. 1e, h**).

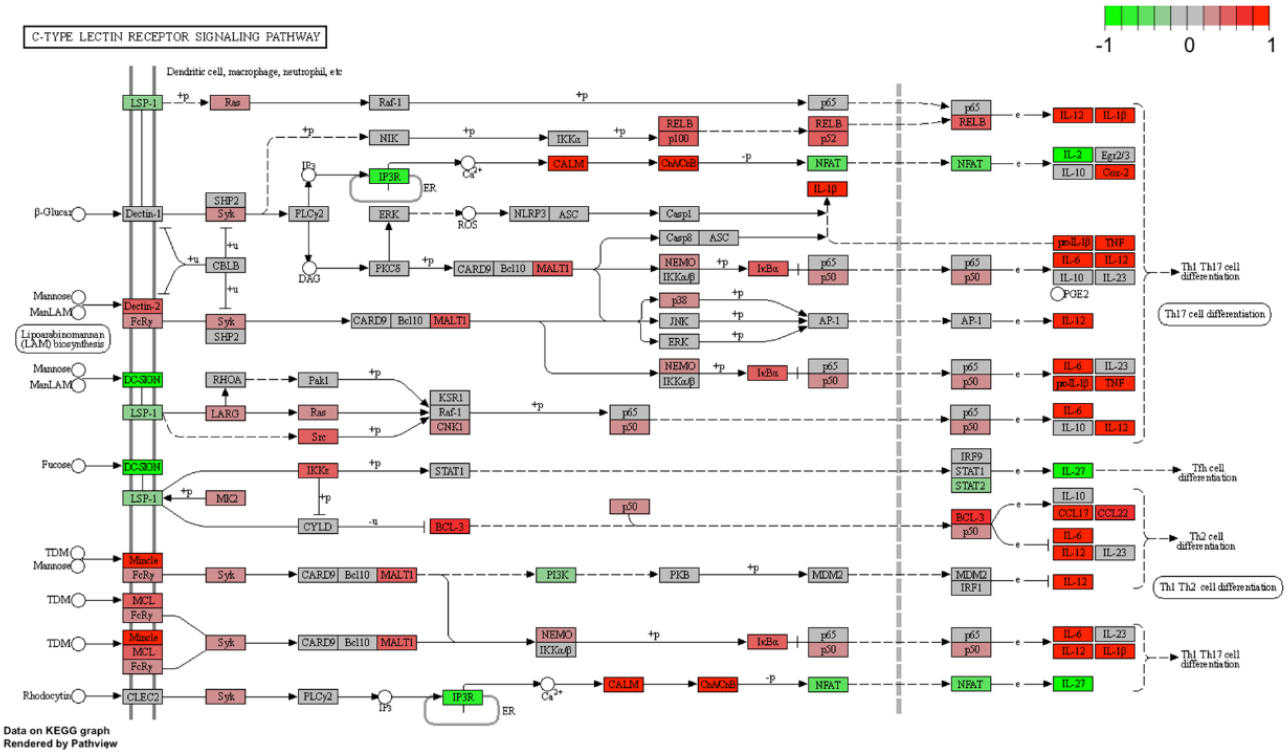
We then studied lipo-circRNA delivery in Balb/c mice using circRNA modified with near-infrared dye IR800, which has great absorption and emission light penetration into tissues, especially superficial lymph nodes. Lipo-circRNA and free circRNA were s.c. injected into the mouse footpad, followed by IVIS imaging and quantification of fluorescence intensities at the same regions of interest including draining popliteal lymph nodes. Impressively, the liposome significantly promoted circRNA delivery to and retention in popliteal lymph nodes for up to 8 days when the experiments ended (**Extended Data Fig. 2a-b**). The long retention of lipo-circRNA in draining lymph nodes is attributed to the efficient delivery by liposomes as well as the excellent stability of circRNA. The efficient circRNA delivery and prolonged circRNA retention are critical to the sustained antigen expression in APCs, which is expected to promote adaptive immunomodulation. We then dissected the intranodal cell distribution of Cy5-labeled lipo-circRNA upon s.c. injection at tail base, followed by lymph node isolation 24 h later, single cell preparation, and flow cytometric analysis of circRNA-Cy5 signal in cells (**Extended Data Fig. 2c-e**). Administration of free circRNA resulted in minimal signal in inguinal draining lymph nodes after 24 h (**Extended Data Fig. 2c**). In contrast, administration of lipo-circRNA led to a significantly higher signal in inguinal draining lymph nodes compared with the free circRNA group (**Extended Data Fig. 2c**). We studied the circRNA distribution kinetics in representative intranodal Ag-presenting DC subsets, including $CD8^+$ T-cell-priming cDCs ($CD11c^+CD11b^-CD8^+$), $CD4^+$ T-cell-priming cDCs ($CD11c^+CD11b^+CD4^+CD103^-CD205^+$), migratory cDCs ($CD11c^+CD11b^-CD8^-CD4^-CD103^+$), and $F4/80^+$ macrophages (**Extended Data Fig. 2e**). Consistent with the above results, Lipo-circRNA promoted cellular uptake of antigen, significantly increasing the frequency of $CD8^+$ T-cell-priming cDCs, $CD4^+$ T-cell-priming cDCs, migratory cDCs, and $F4/80^+$ macrophages. Taken together, these results showed that liposomes efficiently deliver circRNA to draining lymph nodes and promote uptake by APCs in draining lymph nodes, which are prerequisites for the induction of T cell responses.



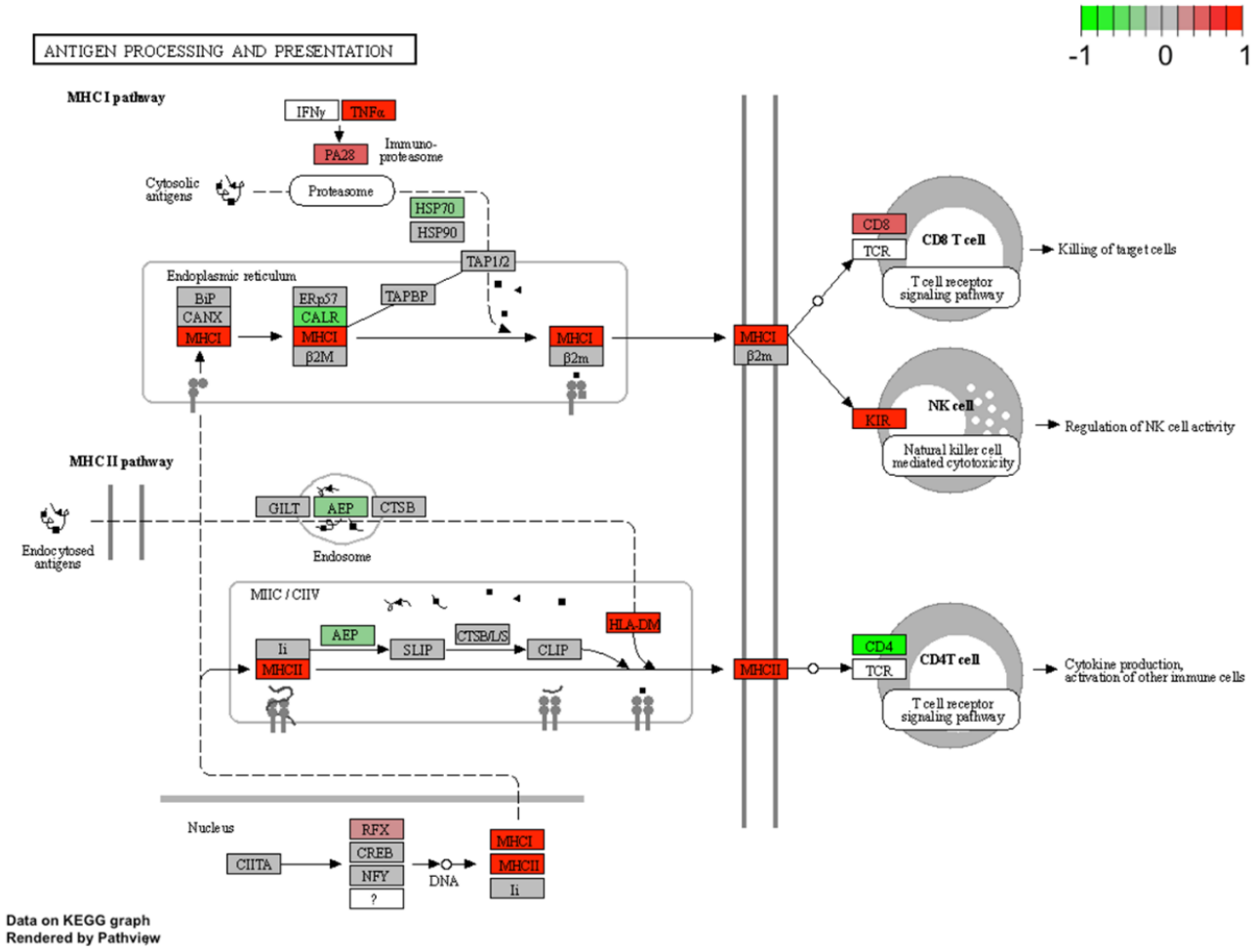

**Supplementary Fig. 7 | Visualize KEGG pathway of TLR signaling pathway between circRNA-SIINFEKL and 5moU-mRNA-OVA by Pathview (https://rdrr.io/bioc/pathview/man/pathview.html).** Gene expression values can be mapped to gradient color scale. Red indicates that circRNA-SIINFEKL induces higher gene expression levels than does 5moU-mRNA-OVA. Green indicates that circRNA-SIINFEKL induces lower gene expression levels than does 5moU-mRNA-OVA.



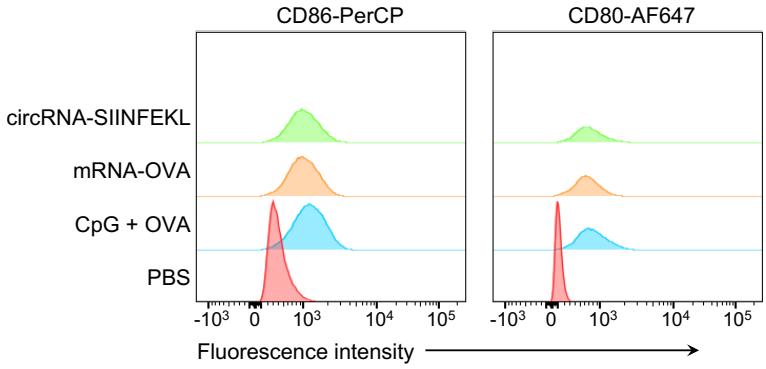

**Supplementary Fig. 8 l Visualize KEGG pathway of RLR signaling pathway between circRNA-SIINFEKL and 5moU-mRNA-OVA by Pathview (https://rdrr.io/bioc/pathview/man/pathview.html).** Gene expression values can be mapped to gradient color scale. Red indicates that circRNA-SIINFEKL induces higher gene expression levels than does 5moU-mRNA-OVA. Green indicates that circRNA-SIINFEKL induces lower gene expression levels than does 5moU-mRNA-OVA.



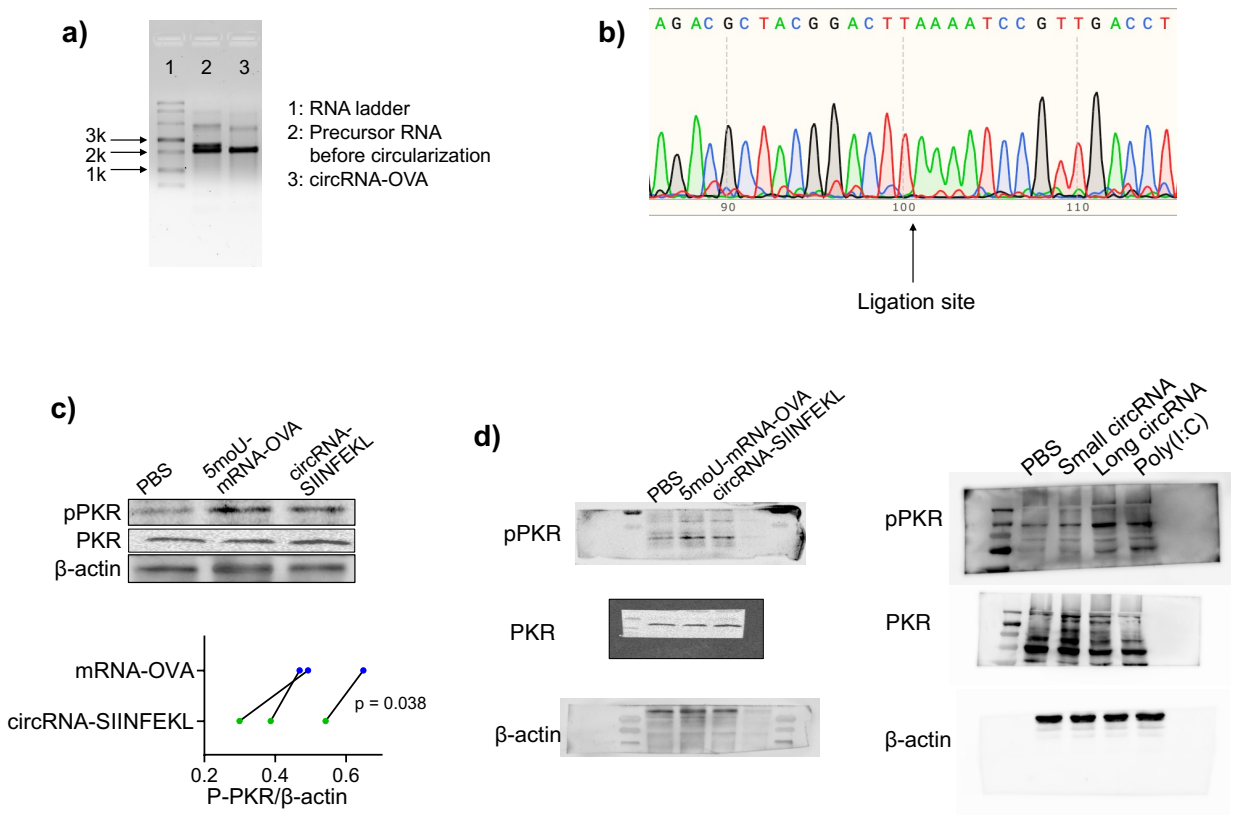

**Supplementary Fig. 9 | Visualize KEGG pathway of CLR signaling pathway between circRNA-SIINFEKL and 5moU-mRNA-OVA by Pathview (https://rdrr.io/bioc/pathview/man/pathview.html).** Gene expression values can be mapped to gradient color scale. Red indicates that circRNA-SIINFEKL induces higher gene expression levels than does 5moU-mRNA-OVA. Green indicates that circRNA-SIINFEKL induces lower gene expression levels than does 5moU-mRNA-OVA.



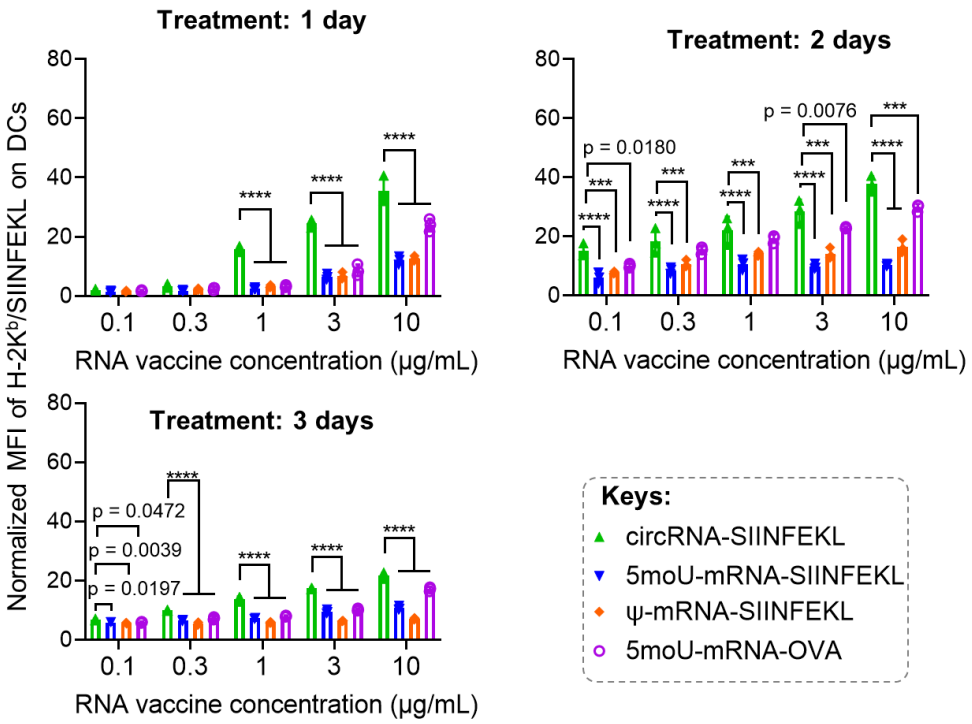

**Supplementary Fig. 10 l Visualization of the KEGG pathway of antigen processing and presentation between circRNA-SIINFEKL and 5moU-mRNA-OVA by Pathview.** Gene expression values can be mapped to gradient color scale. Red indicates that circRNA-SIINFEKL induces higher gene expression levels than does 5moU-mRNA-OVA. Green indicates that circRNA-SIINFEKL induces lower gene expression levels than does 5moU-mRNA-OVA.



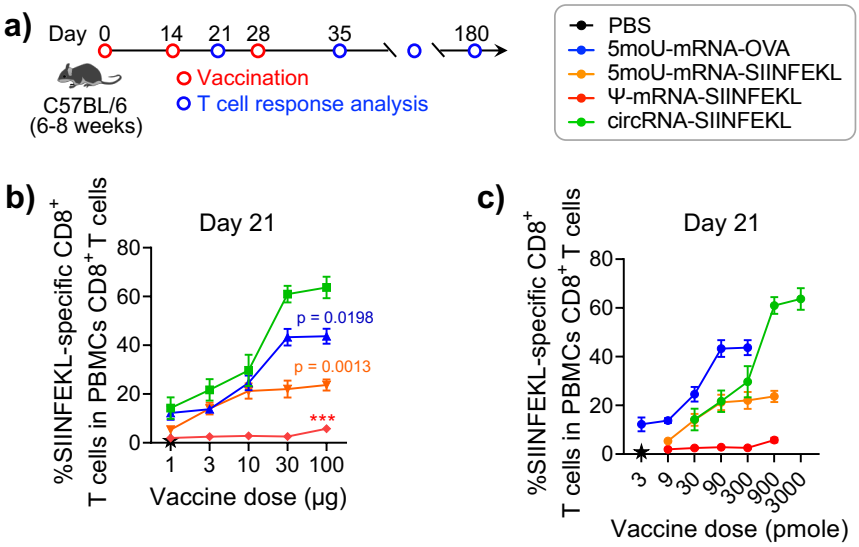

**Supplementary Fig. 11 l circRNA elevated the expression of DC co-stimulatory factors and reduced PKR activation.** DC2.4 cells were treated with PBS, CpG + OVA, 5moU-mRNA-OVA, and circRNA-SIINFEKL (24 h), respectively, followed by flow cytometric analysis. RNA was transfected with lipofectamine 3000.



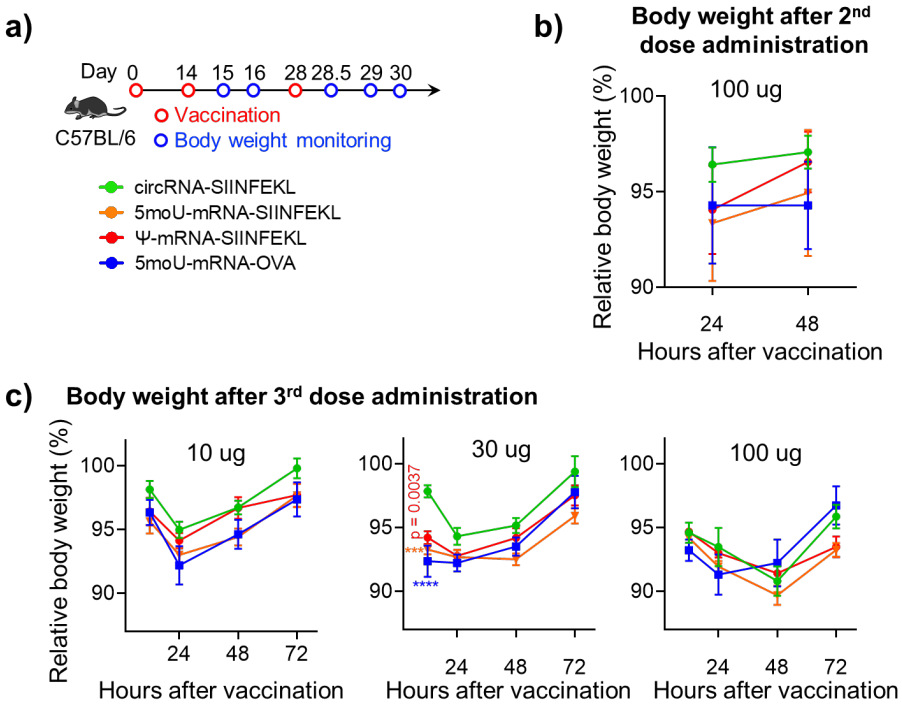

**Supplementary Fig. 12 l circRNA-OVA synthesis and PKR activation by 5moU-mRNA-OVA. a-b**, An agarose gel electrophoresis image (**a**) and their Sanger sequencing results of the cDNA of circRNA-OVA (**b**) showing its successful synthesis. **c**, Western blot analysis of PKR and pPKR, and gel intensity ratio of pPKR over β-actin in DC2.4 cells LNP-transfected with circRNA-SIINFEKL or 5moU-mRNA-OVA (100 nM, 24 h). **p* < 0.05, by paired t-test. **d**, Complete gel images of Fig. 4f and Supplementary Fig. 14b.



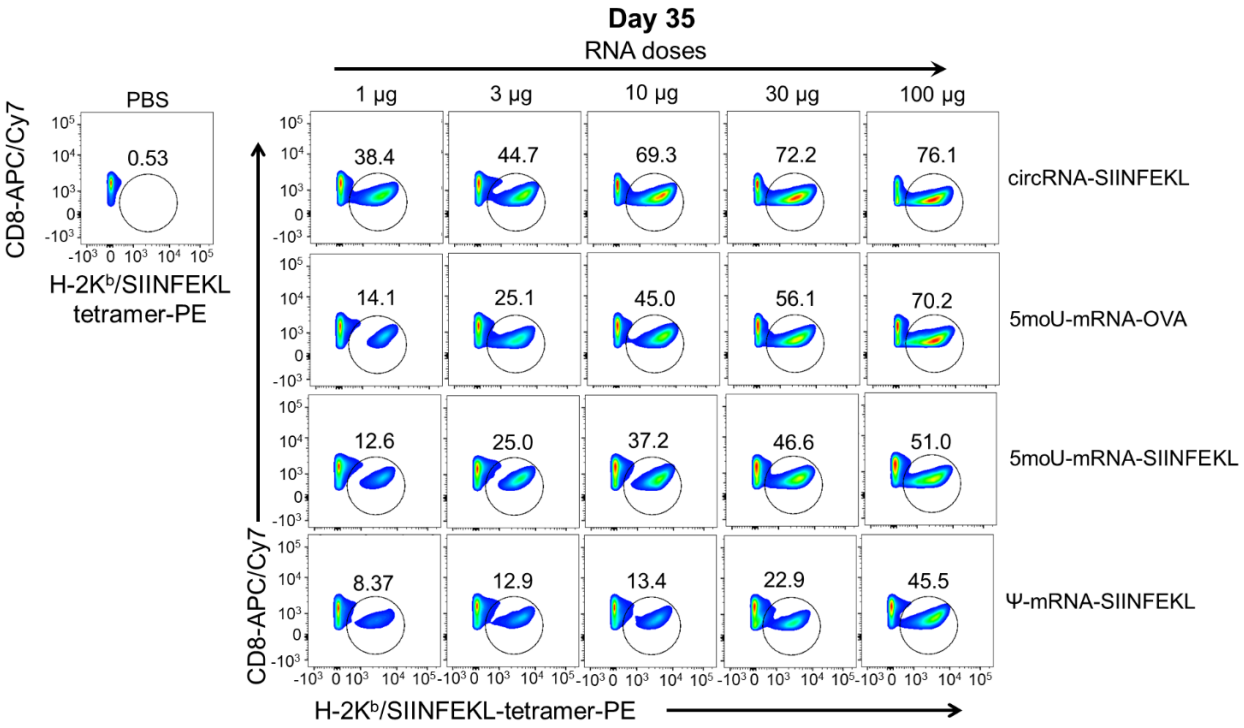

**Supplementary Fig. 13 | Flow cytometric results of SIINFEKL/H-2K$^b$ complex levels on vaccine-treated DC2.4 cells for benchmarking small circRNA-SIINFEKL vaccine against multiple modified mRNA-SIINFEKL and modified mRNA-OVA vaccines.** We compared circRNA-SIINFEKL, 5moU-mRNA-SIINFEKL, Ψ-mRNA-SIINFEKL, and 5moU-mRNA-OVA, all formulated in SM-102 LNPs, for *in vitro* antigen presentation at the corresponding escalating concentrations. All these mRNAs have 5'/3' UTRs, 3' A120, and CleanCap. DC2.4 cells were transfected with vaccines for 1-3 days, followed by flow cytometric analysis of H-2K$^b$/SIINFEKL levels. Ns: non-significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, by one-way ANOVA with Bonferroni post-test.



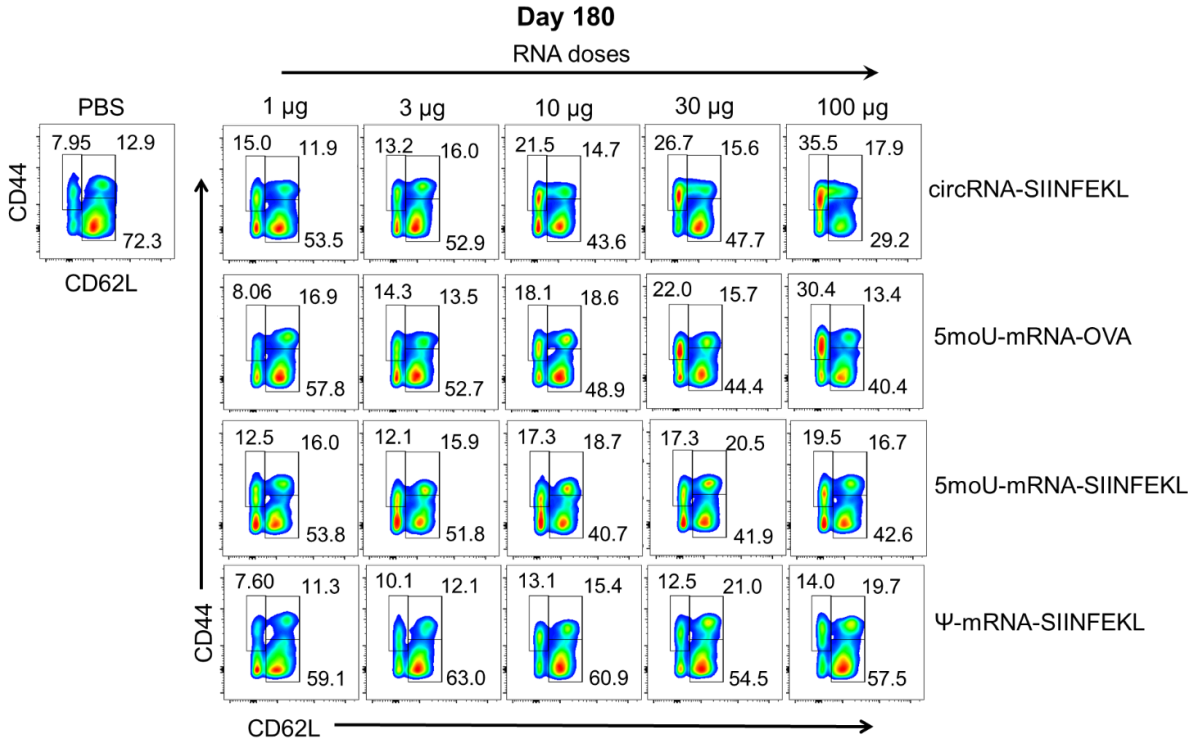

**Supplementary Fig. 14 l Mass dose and molar dose-based comparison of small circRNA vaccines *vs* three modified mRNA vaccines for PBMC SIINFEKL-specific T cells elicited in mice. a,** Study design. C57BL/6 mice (*n* = 4-8; 6-8 weeks) were immunized with circRNA-SIINFEKL and three modified mRNAs at escalating doses, respectively. **b,** Tetramer staining results on day 21 suggest that circRNA-SIINFEKL elicited higher fractions of PBMC SIINFEKL-specific T cells than modified mRNAs at the corresponding doses. Data were the same as Figure 5c). **c,** Molar dose-based comparison of small circRNA vaccines *vs* three modified mRNA vaccines for PBMC SIINFEKL-specific T cells elicited in mice. Data were converted from Figure b). Asterisks: massstatistical significance of the T cell fraction AUC relative to that for circRNA.



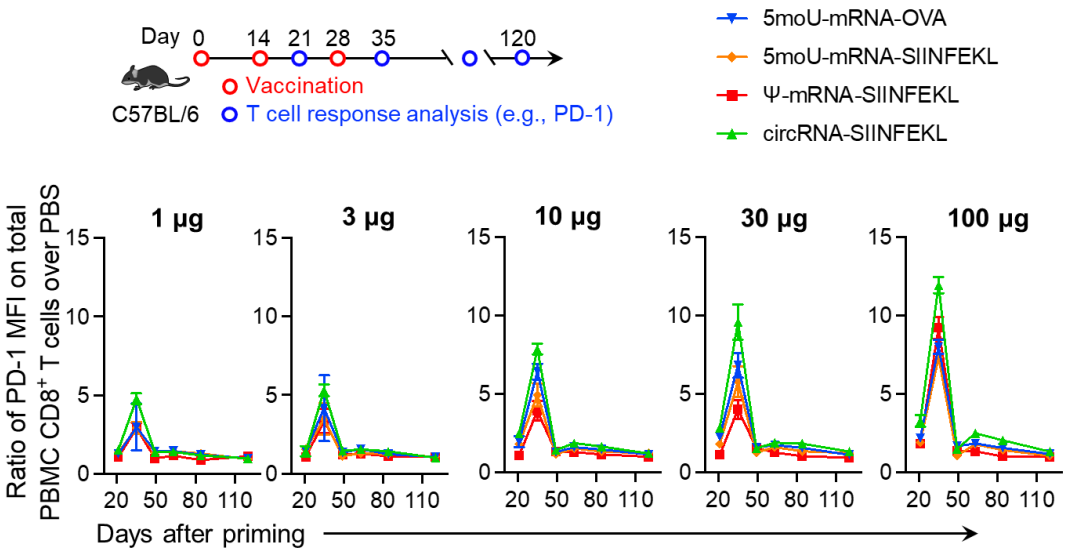

**Supplementary Fig. 15 l Mouse body weight changes after administration with small circRNA-SIINFEKL as well as modified mRNA-SIINFEKL and modified mRNA-OVA vaccines at escalating doses. a**, Study design (**Fig. 5**). All RNAs, including circRNA-SIINFEKL, 5moU-mRNA-SIINFEKL, Ψ-mRNA-SIINFEKL, and 5moU-mRNA-OVA, all separately formulated in SM-102 LNPs. All these mRNAs have 5'/3' UTRs, 3' A120, and CleanCap. C57Bl/c mice (*n*=4-5; 6-8 weeks) were immunized with SM-102 LNPs of circRNA-SIINFEKL and three benchmarking modified mRNA at escalating doses, respectively. **b**, circRNA-SIINFEKL caused less mouse body weight loss than benchmark modified mRNAs at 100 μg dose, as exemplified 1-2 days after the 2nd dose. Data represent mean ± s.e.m. (n = 3-5). **c**, circRNA-SIINFEKL caused less mouse body weight loss and more rapid recovery than benchmark modified mRNAs except for the doses of 100 μg, as exemplified 0.5-3 days after the 3rd dose. Data represent mean ± s.e.m. (n = 5, except for 100 μg-dose groups in which n = 3). Asterisks: statistical significance relative to circRNA. Ns: non-significant, *$p < 0.05$, **$p < 0.01$, by one-way ANOVA with Bonferroni post-test.



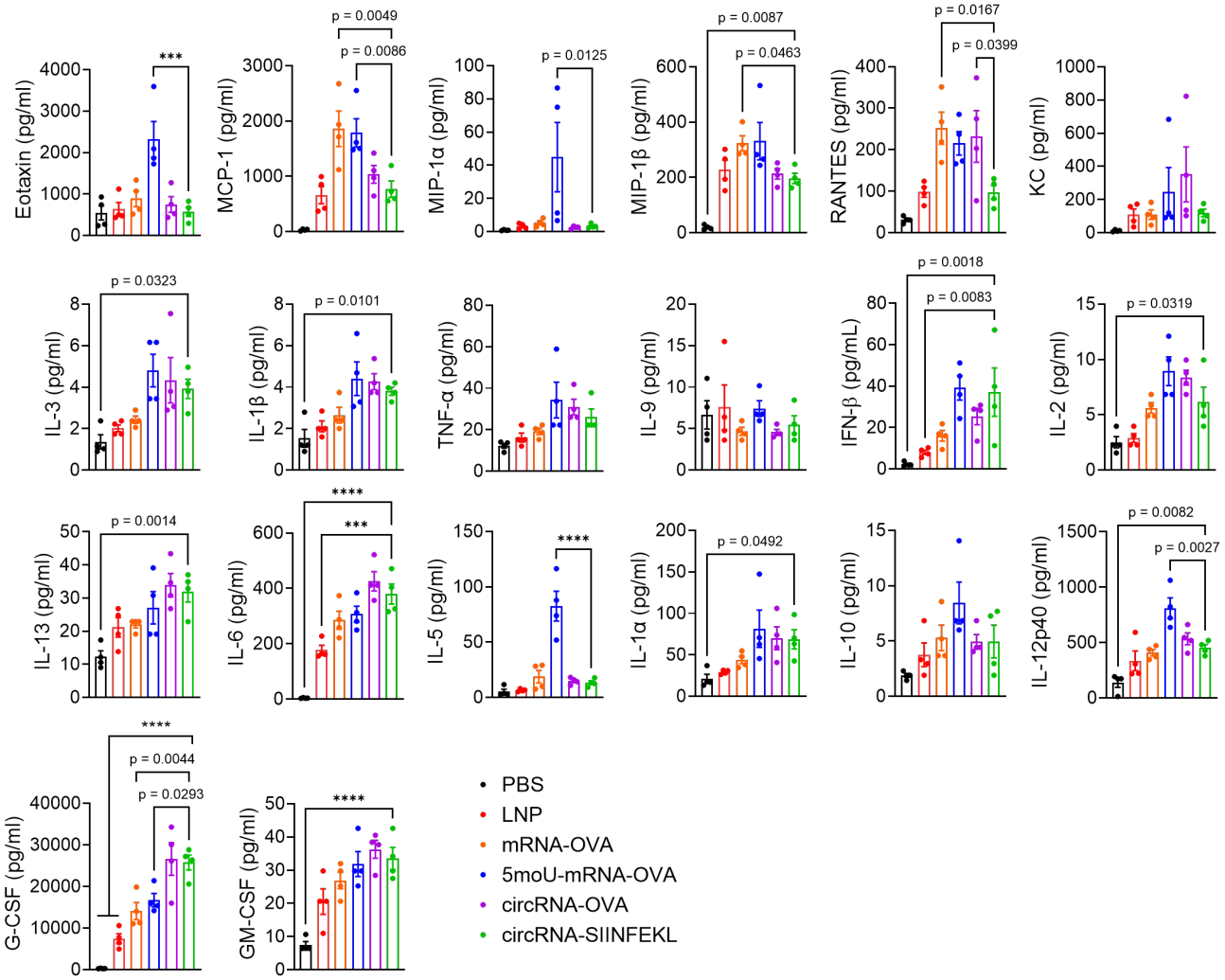

**Supplementary Fig. 16 l** Representative flow cytometry plots for the analysis of PBMC SIINFEKL$^{+}$CD8$^{+}$ T cells on day 35 after three doses of circRNA-SIINFEKL and benchmark modified mRNA at escalating doses (**Fig. 5**), as stained using PE-labeled H-2K$^{b}$-SIINFEKL tetramer.



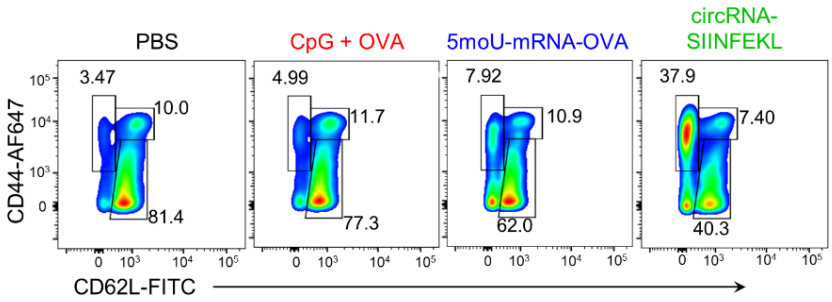

**Supplementary Fig. 17 l** Representative flow cytometry plots for the analysis of PBMC CD8⁺ T cell memory elicited by RNA vaccines at escalating doses (day 180) in mice. The CD8⁺ T cell fractions of effector memory T cells (Tem, CD62L⁻CD44⁺), central memory T cells (Tcm, CD62L$^{high}$CD44⁺), and naive T cells (CD62L⁺CD44⁻) were quantified.



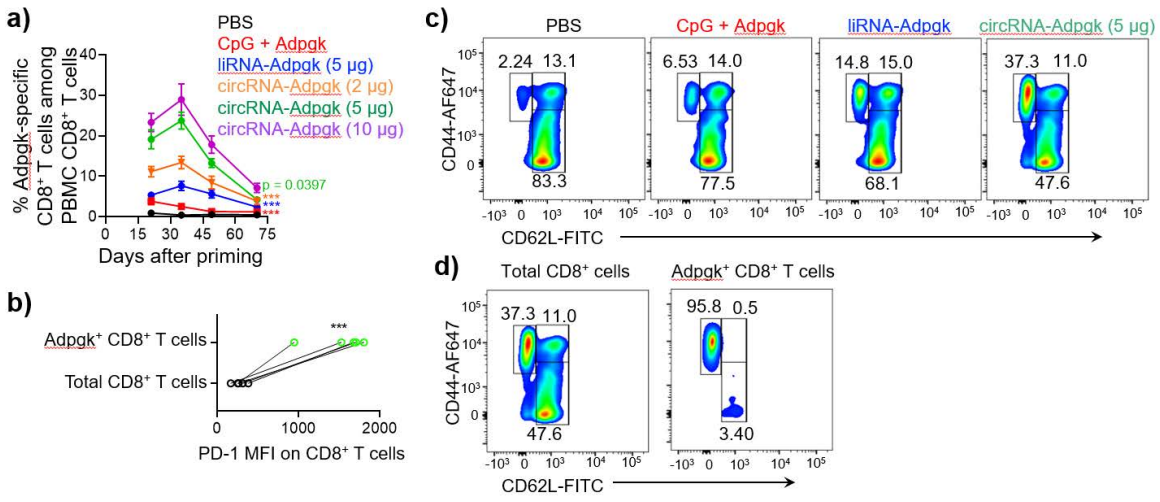

**Supplementary Fig. 18 l** The kinetics of PD-1 expression levels on total PBMC CD8⁺ T cells in mice immunized with circRNA-SIINFEKL and modified mRNA at escalating doses (**Fig. 5**). The results suggest dose-dependent PD-1 upregulation shortly after immunization. The PD-1 expression levels were significantly reduced starting from at most 21 days after the last immunization (day 49).



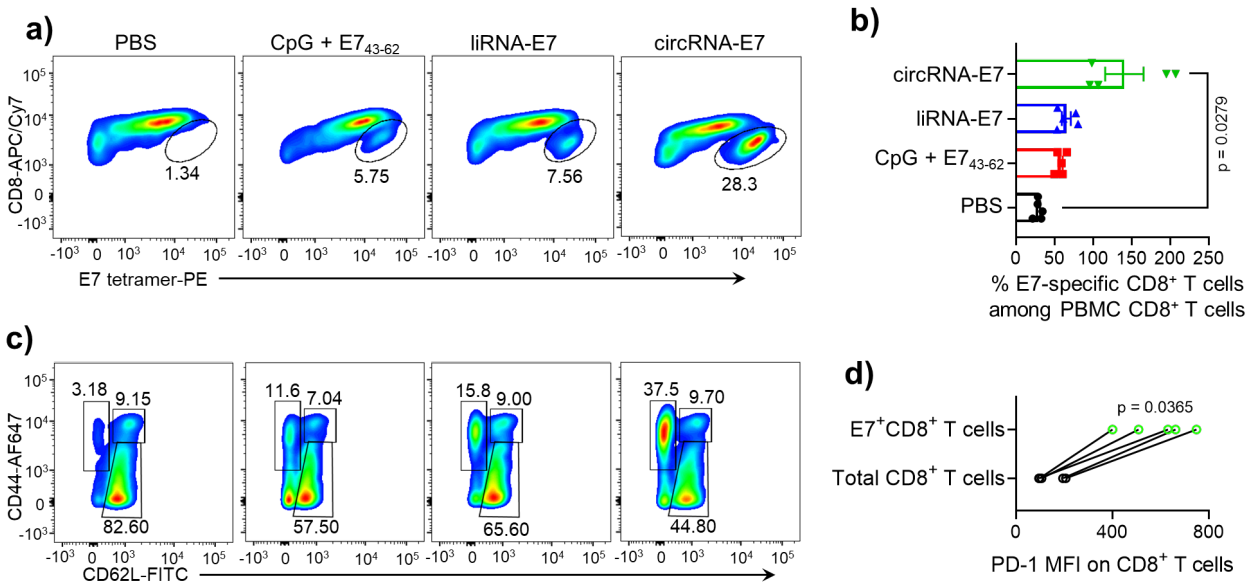

**Supplementary Fig. 19 l** Luminex results of the serum cytokines and chemokines from mice immunized with circRNA-SIINFEKL and benchmark RNA vaccines at a dose of 10 μg (**Fig. 5**). Serum concentrations of chemokines (Eotaxin, RANTES, MCP-1, MIP-1α, MIP-1β, and KC) and cytokines (IL-3, IL-1β, TNF-α, IL-9, IL-6, IFN-β, IL-2, IL-13, IL-5, IL-1α, IL-10, IL-12p40, G-CSF, and GM-CSF) were measured 6 h after the third dose of escalating doses of circRNA-SIINFEKL, mRNA-OVA, 5moU-mRNA-OVA, and circRNA-OVA formulated in SM-102 LNPs ($n = 4$). Mice and ministered with PBS or blank LNPs were used as controls. These data were summarized in **Fig. 5h**. Data: mean ± standard error of the mean (s.e.m.); Asterisks indicate statistically significant differences between circRNA and other groups. $^{**}p < 0.01$, $^{***}p < 0.001$, by one-way ANOVA with Bonferroni post-test. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$, one-way ANOVA with Bonferroni post-test.



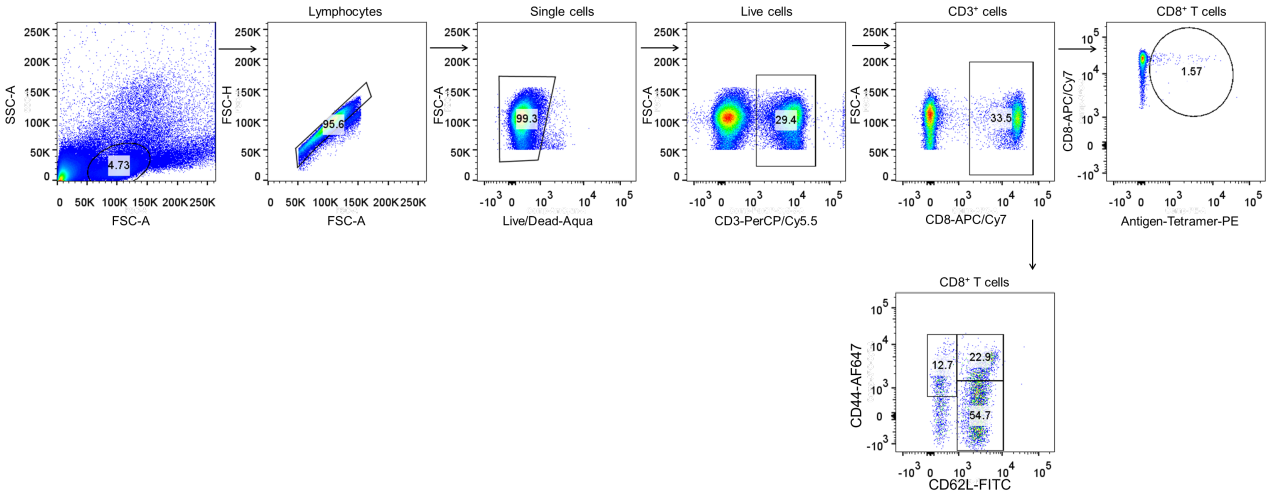

**Supplementary Fig. 20 l** Representative flow cytometry plots showing that circRNA-SIINFEKL (3 x 5 μg on days 0, 14, 28) elicited $CD8^+$ T cell memory (day35) in aged mice (1 year) ($n = 5$).



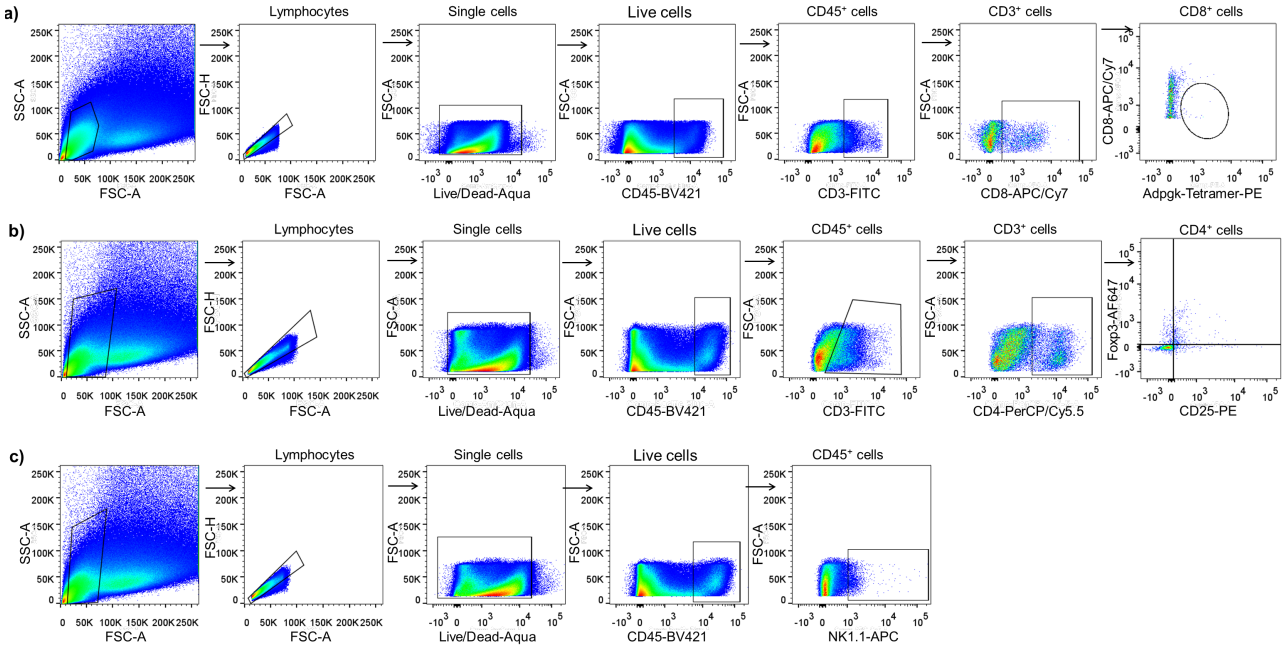

**Supplementary Fig. 21 l Small circRNA-Adpgk elicited potent neoantigen Adpgk-specific T cell responses.** C57BL/6 mice ($n = 5$) were vaccinated on day 0, day 14, and day 28 and PBMCs were analyzed [vaccines were s.c. administered at mouse tail base; circRNA-Adpgk, liRNA-Adpgk (5 μg), and (CpG + Adpgk) (5 μg CpG + 10 μg Adpgk)]. **a**, Tetramer staining results showed the fractions of Adpgk$^+$CD8$^+$ T cells among all live PBMC CD8$^+$ T cells, indicating that circRNA-Adpgk neoantigen vaccine elicited robust and dose-dependent T cell responses. **b**, MFI of PD-1 expression on Adpgk$^+$CD8$^+$ T cells than that on total CD8$^+$ T cells in peripheral blood on day 21 (two-tailed paired t test). **c**, Representative flow cytometry dot plots of effector memory T cells (Tem, CD62L$^-$CD44$^+$), central memory T cells (Tcm, CD62L$^{high}$CD44$^+$), and naive T cells (CD62L$^+$CD44$^-$) in peripheral blood on day 70, showing circRNA-induced T cell memory. **d**, The effector memory T cells on Adpgk$^+$CD8$^+$ T cells than that on total CD8$^+$ T cells in peripheral blood on day 70. Data represent mean ± s.e.m. ($n = 5$). Asterisks in (**a**) indicate statistically significant differences between circRNA and other groups. $^{**}p < 0.01$, $^{***}p < 0.001$, by one-way ANOVA with Bonferroni post-test.



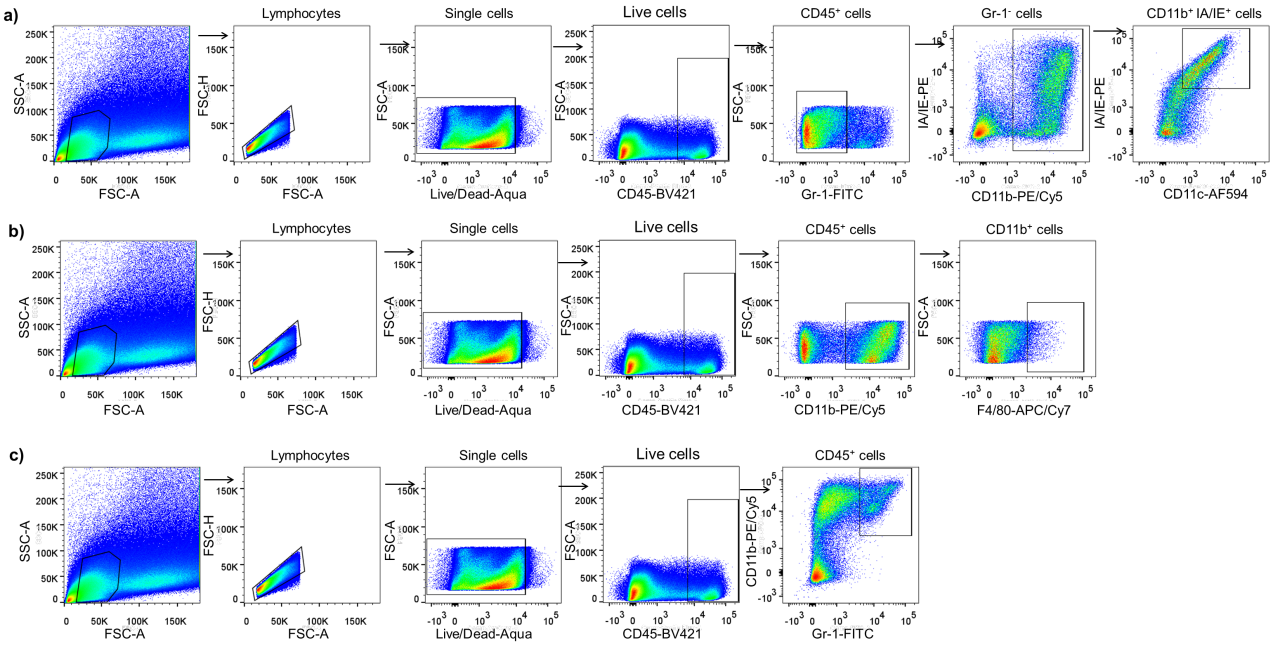

**Supplementary Fig. 22 l Small circRNA-E7 elicited potent HPV E7-specific T cell responses.** C57BL/6 mice ($n = 5$) were vaccinated on day 0, day 14, and day 28, followed by PBMC CD8+ T cell analysis [vaccines were s.c. administered in liposome at mouse tail base; circRNA-E7 (5 μg), liRNA-E7 (5 μg), and (CpG + E7$_{43-62}$) (5 μg CpG + 10 μg E7$_{43-62}$)]. **a-b**, Representative flow cytometry plots (**s**) and frequency (**b**) of PBMC E7+CD8+ T cells on day 21, as stained using PE-labeled H-2D$^b$-RAHYNIVTF tetramer. *$p < 0.05$, **$p < 0.01$, by one-way ANOVA with Bonferroni post-test. **c**, Representative flow cytometry results of PBMC Tem, Tcm, and naive T cells on day 70. **d**, Elevated PD-1 expression on E7+CD8+ T cells relative to that on total PBMC CD8+ T cells on day 21 (**$p < 0.01$, two-tailed paired t test).



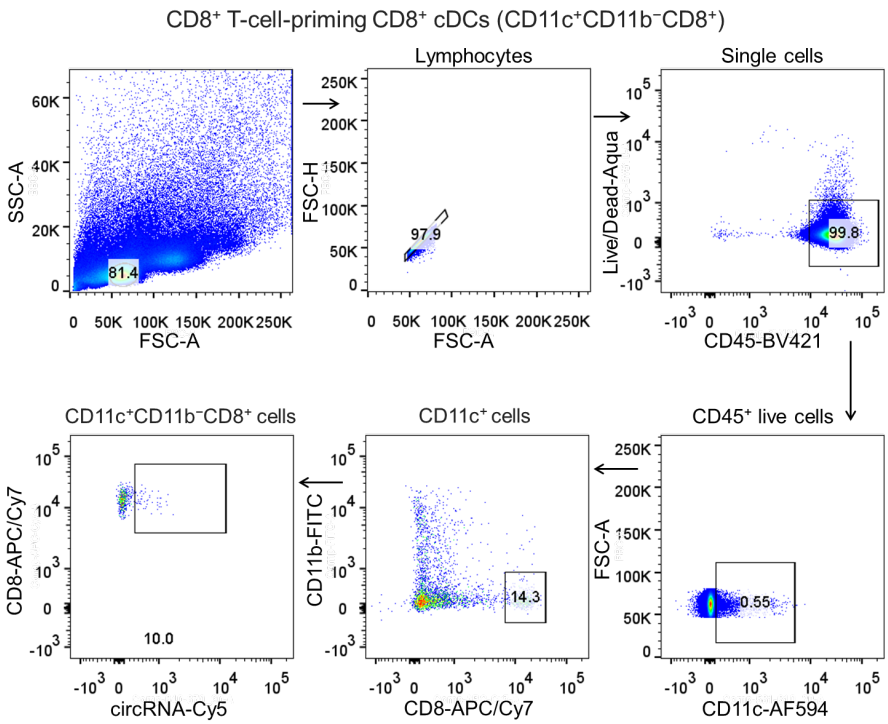

**Supplementary Fig. 23 l** Representative gating tree used in flow cytometric analysis of tetramer staining of PBMC CD8⁺ T cells.



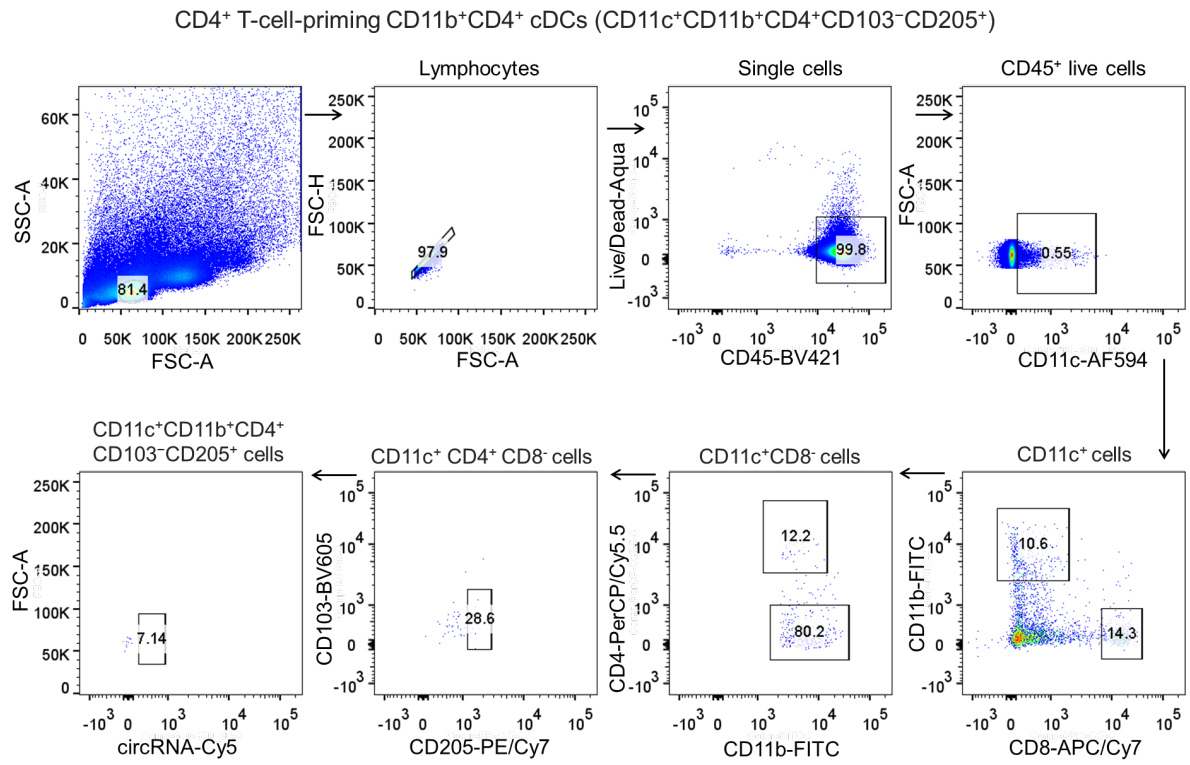

**Supplementary Fig. 24 l** Representative gating tree used in flow cytometric analysis of (**a**) CD8⁺ T cells and Adpgk-specific CD8⁺ T cells, (**b**) CD4⁺ T cells and Treg cells, and (**b**) NK cells in the TME of MC38 tumours.



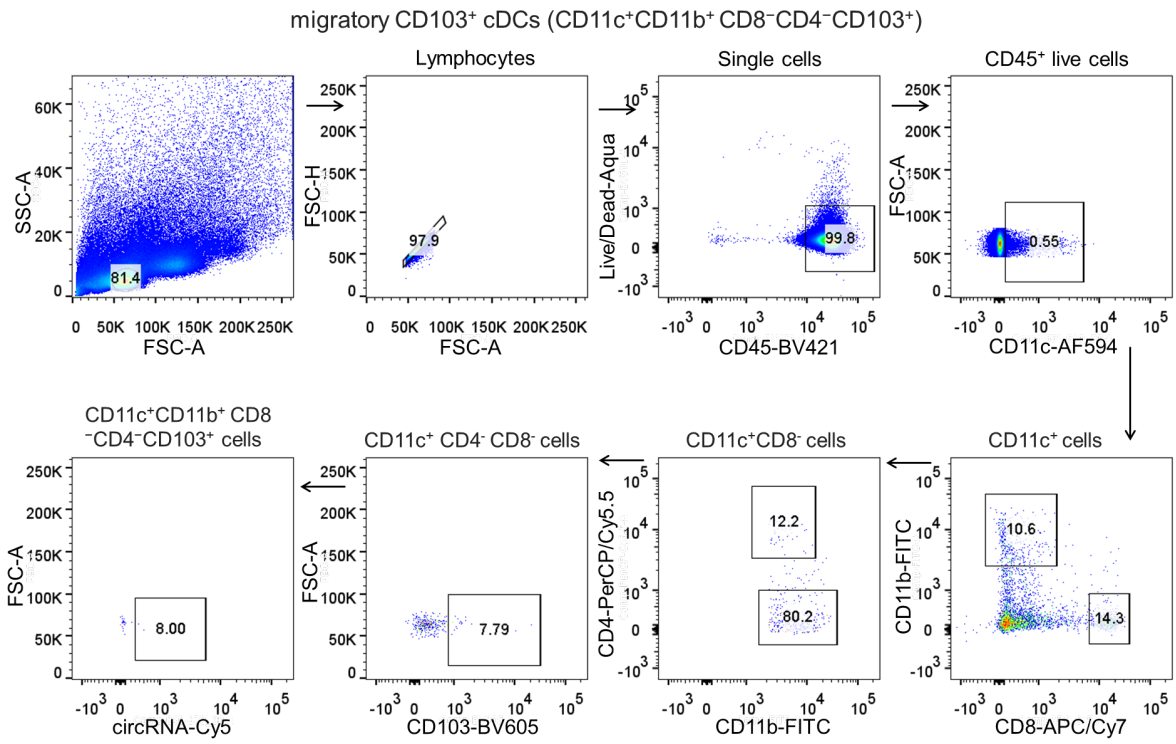

**Supplementary Fig. 25 l** Representative gating tree used in flow cytometric analysis of (**a**) DC cells, (**b**) macrophages, (**c**) and (**c**) MDSC cells in the TME of MC38 tumours.



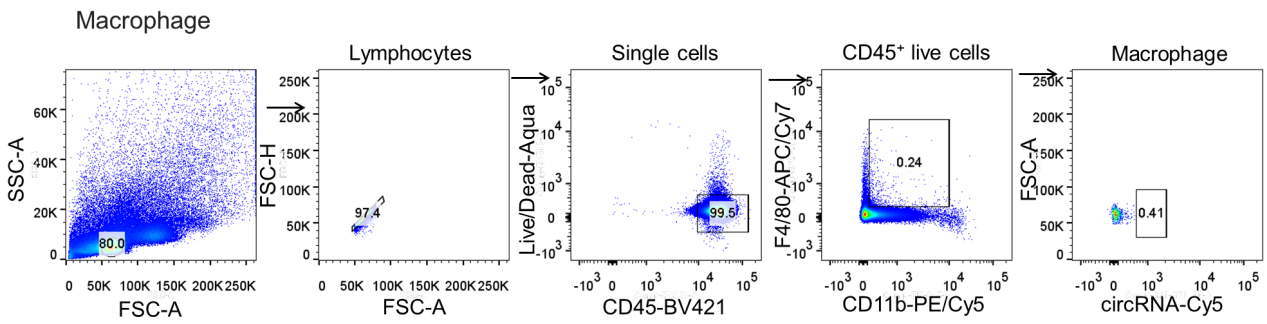

**Supplementary Fig. 26 l** Representative gating tree used in flow cytometric analysis of circRNA⁺ CD8⁺ cDCs in mouse draining lymph nodes.




**Supplementary Fig. 27 l** Representative gating tree used in flow cytometric analysis of circRNA+ CD4+ T-cell-priming CD4+ cDCs in draining lymph nodes.




**Supplementary Fig. 28 l** Representative gating tree used in flow cytometric analysis of circRNA+ migratory CD103+ cDCs in draining lymph nodes.



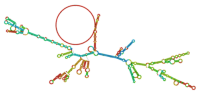
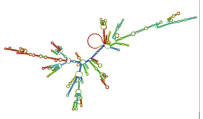

**Supplementary Fig. 29 l** Representative gating tree used in flow cytometric analysis of circRNA+ macrophage in draining lymph nodes.

**Supplementary Table 1 | RNA sequences.**

| Name | Sequence (5' → 3') |
|---|---|
| LINE1 (ORF2-138-86) | CGC AUU AUC UCU CCA CGA AUC CAG CCC UUC AAA GGA UAA UAA CAG AAA AAA AC |
| Rbm3-22NT | UUU AUA AUU UCU UCU UCC AGA A |
| crTMV (IRESmp75) | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG A |
| LINE1-SIINFEKL | CGC AUU AUC UCU CCA CGA AUC CAG CCC UUC AAA GGA UAA UAA CAG AAA AAA ACG GCC ACC AUG GAU AGC AUU AUC AAC UUU GAA AAA CUG |
| crTMV-SIINFEKL | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GAU AGC AUU AUC AAC UUU GAA AAA CUG |
| Rbm3-SIINFEKL | UUU AUA AUU UCU UCU UCC AGA AUC GCC ACC AUG GAG AGU AUA AUC AAC UUU GAA AAA CUG |
| circRNA-ISQ | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GAG AUU AGC CAG GCG GTG CAC GCC GCC CAC GCC GAG ATC AAC GAG GCC GGA AGG |
| circRNA-FLAG | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GAC UAC AAG GAC GAC GAC GAC AAG AUC AUC |
| circRNA-Adpgk | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GCU AGC AUG ACG AAU AUG GAG CUC AUG |
| circRNA-E7 | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GGC CAG GCC GAG CCC GAC AGG GCC CAC UAC AAC AUC GUG ACC UUC UGC UGC AAG UGC GAC |
| circRNA-trp2$_{177-190}$/gp100$_{23-33}$ | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GCC AAC TGC AGC GTG TAC GAT TTC TTC GTG TGG CTG CAC TAC GCC CTG AAA GTG CCT CGT AAC CAG GAT TGG CTG |
| circRNA-trp1$_{103-132}$/trp1$_{455-463}$ | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG GAC AAC TTC AGC GGC CAC AAC TGC GGC ACC TGT AGG CCT GGC TGG AGG GGC GCC GCC TGC AAT CAG AAG ATC CTG ACC GTG AGG AGG AAC TGC ACC GCC CCT GAC AAC CTG GGC TAC ATG TAA |
| circRNA-RBD$_{440-459}$ | UUC GUU UGC UUU UUG UAG UAU AAU UAA AUA UUU GUC AUA UAA GAG AUU GGU UAG AGA UUU GUU CUU UGU UUG AUC GCC ACC AUG AAC CTG GAC AGC AAG GTG GGC GGC AAC TAC AAC TAC CTG TAT AGG CTG TTT AGG AAG AGC |
| circRNA-OVA | AAAATCCGTTGACCTTAAACGGTCGTGTGGGTTCAAGTCCCTCCACCCCCACGCCG GAAACGCAATAGCCGAAAAACAAAAAACAAAAAAAAACAAAAAAAAAAACCAAAAAAAC AAAACACATTAAAACAGCCTGTGGGTTGATCCCACCCACAGGCCCATTGGGCGCTA GCACTCTGGTATCACGGTACCTTTGTGCGCCTGTTTTATACCCCCTCCCCCAACTGT AACTTAGAAGTAACACACACCGATCAACAGTCAGCGTGGCACACCAGCCACGTTTT GATCAAGCACTTCTGTTACCCCGGACTGAGTATCAATAGACTGCTCACGCGGTTGA AGGAGAAAGCGTTCGTTATCCGGCCAACTACTTCGAAAAACCTAGTAACACCGTGG AAGTTGCAGAGTGTTTCGCTCAGCACTACCCCAGTGTAGATCAGGTCGATGAGTCA CCGCATTCCCCACGGGCGACCGTGGCGGTGGCTGCGTTGGCGGCCTGCCCATGG GGAAACCCATGGGACGCTCTAATACAGACATGGTGCGAAGAGTCTATTGAGCTAGT TGGTAGTCCTCCGGCCCCTGAATGCGGCTAATCCTAACTGCGGAGCACACACCCTC AAGCCAGAGGGCAGTGTGTCGTAACGGGCAACTCTGCAGCGGAACCGACTACTTT GGGTGTCCGTGTTTCATTTTATTCCTATACTGGCTGCTTATGGTGACAATTGAGAGA TCGTTACCATATAGCTATTGGATTGGCCATCCGGTGACTAATAGAGCTATTATATATC CCTTTGTTGGGTTTATACCACTTAGCTTGAAAGAGGTTAAAACATTACAATTCATTGT TAAGTTGAATACAGCAAAGCCACCATGGGCAGCATCGGCGCCGCCAGCATGGAGTT CTGCTTCGACGTGTTCAAGGAGCTGAAGGTGCACCACGCCAACGAGAACATCTTCT ACTGCCCCATCGCCATCATGAGCGCCCTGGCCATGGTGTACCTGGGCGCCAAGGA CAGCACCCGGACCCAGATCAACAAGGTGGTGCGGTTCGACAAGCTGCCCGGCTTC GGCGACAGCATCGAGGCCCAGTGCGGCACCAGCGTGAACGTGCACAGCAGCCTG |

| | |
|---|---|
| | CGGGACATCCTGAACCAGATCACCAAGCCCAACGACGTGTACAGCTTCAGCCTGGC<br>CAGCCGGCTGTACGCCGAGGAGCGGTACCCCATCCTGCCCGAGTACCTGCAGTGC<br>GTGAAGGAGCTGTACCGGGGCGGCCTGGAGCCCATCAACTTCCAGACCGCCGCCG<br>ACCAGGCCCGGGAGCTGATCAACAGCTGGGTGGAGAGCCAGACCAACGGCATCAT<br>CCGGAACGTGCTGCAGCCCAGCAGCGTGGACAGCCAGACCGCCATGGTGCTGGT<br>GAACGCCATCGTGTTCAAGGGCCTGTGGGAGAAGACCTTCAAGGACGAGGACACC<br>CAGGCCATGCCCTTCCGGGTGACCGAGCAGGAGAGCAAGCCCGTGCAGATGATGT<br>ACCAGATCGGCCTGTTCCGGGTGGCCAGCATGGCCAGCGAGAAGATGAAGATCCT<br>GGAGCTGCCCTTCGCCAGCGGCACCATGAGCATGCTGGTGCTGCTGCCCGACGAG<br>GTGAGCGGCCTGGAGCAGCTGGAGAGCATCATCAACTTCGAGAAGCTGACCGAGT<br>GGACCAGCAGCAACGTGATGGAGGAGCGGAAGATCAAGGTGTACCTGCCCCGGAT<br>GAAGATGGAGGAGAAGTACAACCTGACCAGCGTGCTGATGGCCATGGGCATCACC<br>GACGTGTTCAGCAGCAGCGCCAACCTGAGCGGCATCAGCAGCGCCGAGAGCCTGA<br>AGATCAGCCAGGCCGTGCACGCCGCCCACGCCGAGATCAACGAGGCCGGCCGGG<br>AGGTGGTGGGCAGCGCCGAGGCCGGCGTGGACGCCGCCAGCGTGAGCGAGGAG<br>TTCCGGGCCGACCACCCCTTCCTGTTCTGCATCAAGCACATCGCCACCAACGCCGT<br>GCTGTTCTTCGGCCGGTGCGTGAGCCCCTGAAAAAAACAAAAAACAAAACGGCTAT<br>TATGCGTTACCGGCGAGACGCTACGGACTT |
| circRNA-RBD | CAAAUCCGUUGACCUUAAACGGUCGUGUGGGUUCAAGUCCCUCCACCCCCACGC<br>CGGAAACGCAAUAGCCGAAAAACAAAAAACAAAAAAAACAAAAAAAAAACCAAAAAA<br>ACAAAACACAUUAAAACAGCGGAUGGGUACCCCACCAUCCGACCCACUGGGUGUA<br>GUACUCUGGUACUUCGUACCUUUGUACGCCUGUUCUUCCCAUUGUACCCUUCCU<br>GAACUUCCAACCCAAGUAACGUUAGAAGCUCAACAUUUAGUACAACAGGAAGCAC<br>CACAUCCAGUGGUGUUUAGUACAAGCACUUCUGUUUCCCCGGAGCGAGGUAUAG<br>GCUGUACCCACUGCCAAAAACCUUUAACCGUUAUCCGCCAACCAACUACGUAAAA<br>GCUAGUAGUAUUAUGUUUUUAACUAGGCGUUCGAUCAGGUGGAUUUCCCCUCCA<br>CUAGUUUGGUCGAUGAGGCUAGGAAUUCCCCACGGGUGACCGUGUCCUAGCCUG<br>CGUGGCGGCCAACCCAGCCCACUCACUAUUUGUUUUCGCGCCCAGUUGCAAAAA<br>GUGUCGGGGCUGGGACGCCUUUUUAUAGACAUGGUGUGAAGACUCGCAUGUGC<br>UUGGUUGUGAUUCCUCCGGCCCCUGAAUGCGGCUAACCUUAACCCUGGAGCCUU<br>GUGUCACAAACCAGUGAUGAUAAGGUCGUAAUGAGCAAUUCCGGGACGGGACCG<br>ACUACUUUGGGUGUCCGUGUUUCUUAUUUUUCUUAUUAUUGUCUUAUGGUCACA<br>GCAUAUAUAUAACAUAUACUGUGAUCACUAGUGCCACCAUGGAUGCAAUGAAGAG<br>AGGGCUCUGCUGUGUGCUGCUGCUGUGUGGAGCAGUCUUCGUUUCGCCCAGCC<br>AGGAAAUCCAUGCCCGAUUCAGAAGACGCGUCCAGCCAACCGAGAGCAUCGUCA<br>GAUUUCCCAACAUUACAAAUCUGUGCCCCUUCGACGAGGUGUUCAACGCCACAC<br>GCUUCGCUUCAGUGUACGCAUGGAACCGCAAGCGCAUAUCUAACUGCGUCGCGG<br>AUUAUUCUGUCCUCUACAACCUCGCCCCUUUCUUCACCUUCAAGUGCUACGGAG<br>UGUCACCGACUAAGCUGAACGAUCUCUGCUUUACCAACGUCUACGCGGACUCCU<br>UCGUGAUAAGAGGUGAUGAAGUGAGACAAAUAGCCCCAGGUCAGACUGGUAACA<br>UCGCAGAUUACAACUACAAAUUGCCUGAUGAUUUCACUGGUUGCGUUAUCGCGU<br>GGAACUCUAAUAAGCUCGAUUCUAAGGUCAGUGGUAACUACAAUUACCUGUACCG<br>CUUGUUUAGGAAGUCAAACCUGAAGCCUUUCGAGAGGGAUAUUUCAACCGAAAUC<br>UAUCAAGCGGGUAACAAGCCGUGUAACGGUGUGGCAGGAUUUAACUGCUACUUC<br>CCCCUGCGGUCUUACAGCUUCCGGCCAACCUAUGGCGUGGGUCACCAACCUUAU<br>CGCGUGGUGGUUCUGAGUUUCGAACUGUUGCACGCUCCCGCCACGGUAUGCGG<br>UCCCAAGAAGAGCACUAACUUGGUGAAGAAUAAGUGCGUGAAUUUCGGCGGAGG<br>AGGCAGCGGCGGAGGAGGCAGCGGAAGCGGCUACAUCCCAGAAGCCCCUAGAGA<br>CGGACAGGCUUACGUGCGAAAAGACGGCGAGUGGGUGCUGCUGAGCACAUUCCU<br>GGGAAGGAGCUGAUGAUGAACCGGUAAAAAACAAAAAACAAAACGGCUAUUAUGC<br>GUUACCGGCGAGACGCUACGGACUU |

**Supplementary Table 2 | Comparison of the structures of circRNA-SIINFEKL, circRNA-OVA, and three modified mRNA encoding OVA or SIINFEKL.** The secondary structures were predicted using RNAfold.

| | circRNA-SIINFEKL | **5moU**-mRNA-OVA | mRNA-OVA | circRNA-OVA | **5moU**-mRNA-SIINFEKL | **Ψ**-mRNA-SIINFEKL |
|---|---|---|---|---|---|---|
| Predicted secondary structures<br>0 — 1<br>Probability |  |  |  |  |  |  |
| Structural modifications | • IRES<br>• Kozak sequence | • 5′/3′ UTRs<br>• 5′ artificial cap<br>• 3′ A120 | • 5′/3′ UTRs<br>• 5′ artificial cap<br>• 3′ A120 | • IRES<br>• Kozak sequence<br>• Spacer | • 5′/3′ UTRs<br>• 5′ artificial cap<br>• 3′ A120 | • 5′/3′ UTRs<br>• 5′ artificial cap<br>• 3′ A120 |
| Nucleotide modifications | None |  <br>5-Methoxyuridine (5moU) | None | None |  <br>5moU |  <br>Pseudouridine (Ψ) |

**Supplementary Table 3 | A list of antibodies used in this study.**

| Targets | Fluorochromes | Clones | Vendors | Catalogue # |
|---|---|---|---|---|
| CD45 | Brilliant Violet 421 | 30-F11 | BioLegend | 103133 |
| CD11c | Alexa Fluor 594 | N418 | BioLegend | 117346 |
| CD11b | FITC | M1/70 | BioLegend | 101205 |
| CD8a | APC/Cy7 | 53-6.7 | BioLegend | 100713 |
| CD4 | PerCP/Cy5.5 | GK1.5 | BioLegend | 100433 |
| CD103 | Brilliant Violet 605 | 2E7 | BioLegend | 121433 |
| CD205 | PE/Cy7 | NLDC-145 | BioLegend | 138209 |
| F4/80 | APC/Cy7 | BM8 | BioLegend | 123117 |
| CD11b | PE/Cy5 | M1/70 | BioLegend | 101209 |
| NK1.1 | APC | S17016D | BioLegend | 156505 |
| CD3 | PerCP/Cy5.5 | 17A2 | BioLegend | 100217 |
| CD279(PD-1) | Brilliant Violet 421 | 29F.1A12 | BioLegend | 135217 |
| IFN-γ | PE | XMG1.2 | BioLegend | 505807 |
| TNF-α | FITC | MP6-XT22 | BioLegend | 506303 |
| CD44 | PE/Cy5 | IM7 | BioLegend | 103009 |
| CD62L | FITC | MEL-14 | BioLegend | 104405 |
| CD25 | FITC | 3C7 | BioLegend | 101907 |
| FoxP3 | Alexa Fluor 647 | MF-14 | BioLegend | 126407 |
| CD279(PD-1) | N/A | RMP1-14-CP162 | Bio X Cell | CP162 |
| CD152(CTLA-4) | N/A | 9H10-CP146 | Bio X Cell | CP146 |
| CD4 | N/A | YTS 177 | Bio X Cell | BP0003-3 |
| CD8α | N/A | YTS 169.4 | Bio X Cell | BP0117 |
| NK1.1 | N/A | PK136 | Bio X Cell | BP0036 |
| Phospho-PKR (Thr446) | N/A | | Thermo Fisher | PA5-121356 |
| EIF2AK2/PKR | N/A | | Thermo Fisher | 18244-1-AP |
| FLAG M2 | N/A | | Sigma-Aldrich | F1804 |
| $\boldsymbol{\beta}$-actin | N/A | 15G5A11/E2 | Thermo Fisher | MA1-140 |

**Supplementary Table 4 | A list of peptides used in this study.**

| **Peptide names** | **Sequences** |
|---|---|
| SIINFEKL | SIINFEKL |
| ISQ | ISQAVHAAHAEINEAGR |
| FLAG | DYKDDDDK |
| Adpgk | ASMTNMELM |
| HPV-16 $E7_{43-62}$ | GQAEPDRAHYNIVTFCCKCD |
| SARS CoV-2 Spike protein $RBD_{440-459}$ | NLDSKVGGNYNYLYRLFRKS |
| $Trp2_{177-190}$ | ANCSVYDFFVWLHY |
| $gp100_{23-33}$ | ALKVPRNQDWL |
| $Trp1_{103-132}$ | DNFSGHNCGTCRPGWRGAACNQKILTVRN |
| $Trp1_{455-463}$ | CTAPDNLGYM |

**Supplementary reference**

Zhuang, Y. *et al.* PEGylated cationic liposomes robustly augment vaccine-induced immune responses: Role of lymphatic trafficking and biodistribution. *J Control Release* **159**, 135-14 (2012).